

# Combinatorial Algorithms (Math 482), 2026 Spring      Problem Set 3

Due by Wednesday, February 11, 2026.

1. **Substitution Method.** Use the substitution method for proving the following bounds. For the base cases, assume  $T(c) = 1$  for small constants  $c$ .

(a) If  $T(n) = T(\frac{4n}{7}) + T(\frac{n}{3}) + \Theta(n)$ , then  $T(n) = \Theta(n)$ .

(b) If  $T(n) = 2T(\frac{3n}{7}) + T(\frac{n}{7}) + \Theta(n)$ , then  $T(n) = \Theta(n \log n)$ .

2. **Master Method.** Find tight asymptotic bounds for the following recurrences using the master theorem. For the boundary cases, assume  $T(c) = 1$  for small constants  $c$ .

(a)  $T(n) = 9T(\frac{n}{3}) + n^2$ ,

(b)  $T(n) = 2T(\frac{2n}{3}) + n^{4/3}$ ,

(c)  $T(n) = 27T(\frac{n}{3}) + n^3 \log n$ ,

(d)  $T(n) = 5T(\frac{n}{2}) + n^2 \log \log n$ .

3. **Recursion Tree Method.** Use the recursion tree method to solve the following recurrences. For the base cases, assume  $T(c) = 1$  for small constants  $c$ .

(a)  $T(n) = 2T(n-3) + O(1)$ .

(b)  $T(n) = 3T(\frac{n}{3}) + \frac{n}{\log n}$ ,

In each case, answer the following questions:

- (i) What is the height of the recursion tree?
- (ii) What is the amount of work done on each level?
- (iii) Calculate the summation over all levels.